EXECUTIVE SUMMARY

This report analyses the sales data of the company to understand the current revenue structure, identify potential business risks, and highlight areas for growth. The analysis reveals that revenue is heavily concentrated in the North and West regions, driven by a higher number of stores and higher average sales per store. Furthermore, significant performance gaps exist between stores operating in the same region and size category. Key recommendations include utilizing top-performing stores as internal benchmarks to improve weaker stores and investigating the structural challenges in the East and South regions.

1. INTRODUCTION & BUSINESS PROBLEM

The management has requested an analysis of the company's sales data to better understand the overall revenue structure and identify potential business risks and opportunities. A specific concern is how dependent the total revenue is on specific regions or customer segments, and where the company should focus its account management and sales efforts.

To address this, a descriptive and exploratory data analysis was conducted on a dataset containing information on store locations, sizes, employee counts, customer footfall, and monthly sales. The objective of this report is to present the main findings of this analysis and provide evidence-based managerial recommendations.

The dataset initially contains information from 2,485 stores. As is common with raw data, there are some missing values. Figure 1 provides an overview of the data and highlights these missing values.

	Missing	Missing %	Observations	Minimum	Median	Maximum	Mean	Std Dev
Store Area (sqm)	124	5.0	2361	502	2337	18790	2465.8	1746.0
Number of Employees	74	3.0	2411	6	28	49	27.8	10.6
Customer Footfall	99	4.0	2386	109	1220	14228	1458.5	1115.7
Monthly Sales (€)	0	0.0	2485	862	13688	279144	17735.1	17486.2

Figure 1 Overview of All of the Sales data.

While there is no missing data for monthly sales, 124 values are missing in the Store Area, 74 values missing in the Employees and 99 Customer Footfall category. To ensure a reliable analysis, the stores with missing data were removed, leaving a total of 2,199 stores in the dataset. Since the missing data was distributed evenly across the regions, this exclusion should not significantly skew the results, though it is worth keeping in mind when making decisions.

Among the remaining 2,199 stores, there is considerable variance between the minimum and maximum values. For instance, the monthly Customer Footfall ranges widely from 109 to 14,228. To identify trends more reliably, the data was divided into different segments. The stores are in four regions: East, North, South, and West, which makes it easier to find patterns and differences when comparing locations.

Additionally, the stores were segmented by Store Area into three categories: Small, Medium, and Large. The 'Store Area' variable was used as a bin: Small stores have an area of less than 1,750, Medium stores are between 1,750 and 3,500, and Large stores are over 3,500.

Region	Store Count	Store Area	Number of Employees	Customer Footfall	Monthly Sales
East	460	2370.0	27.1	1325.6	13822.2
North	678	2509.3	28.5	1507.7	20433.1
South	419	2532.7	27.2	1407.9	14287.8
West	642	2420.0	27.9	1516.4	19776.2

Figure 2 Data after cleaning and Region averages.

As shown in Figure 2, a larger concentration of stores is in the North and West regions. When looking at the regional averages, there are no major differences in the Store Area or the number of Employees. However, in terms of Customer Footfall, the East and South regions are slightly behind the North and West. The most significant difference is seen in Monthly Sales, where the North region generates over €6,000 more in average monthly sales compared to the others.

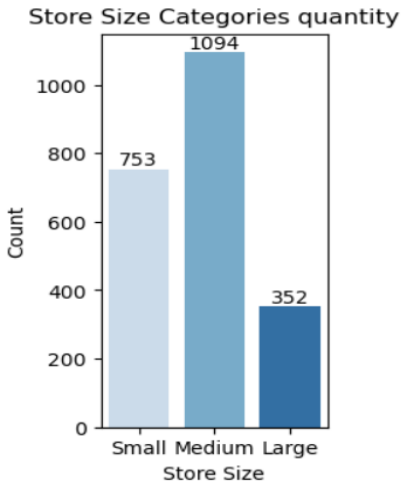

Figure 3 Store Size categories quantity.

Figure 3 represents the quantity of stores across the different size categories, allowing for a more balanced comparison between them.

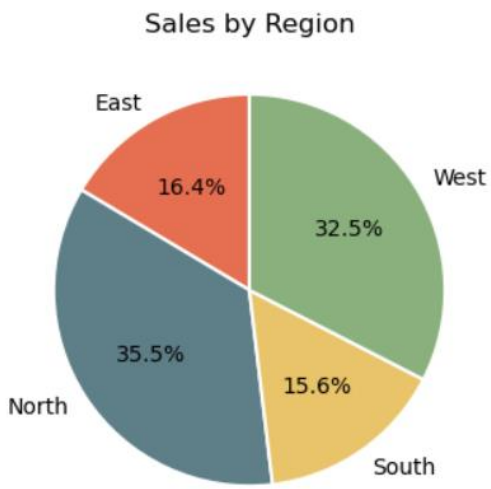

Figure 4 Total Sales Breakdown by Region

According to Figure 4, the total sales for the month were heavily concentrated in the West and North regions. This can be explained by the higher number of stores located in these areas.

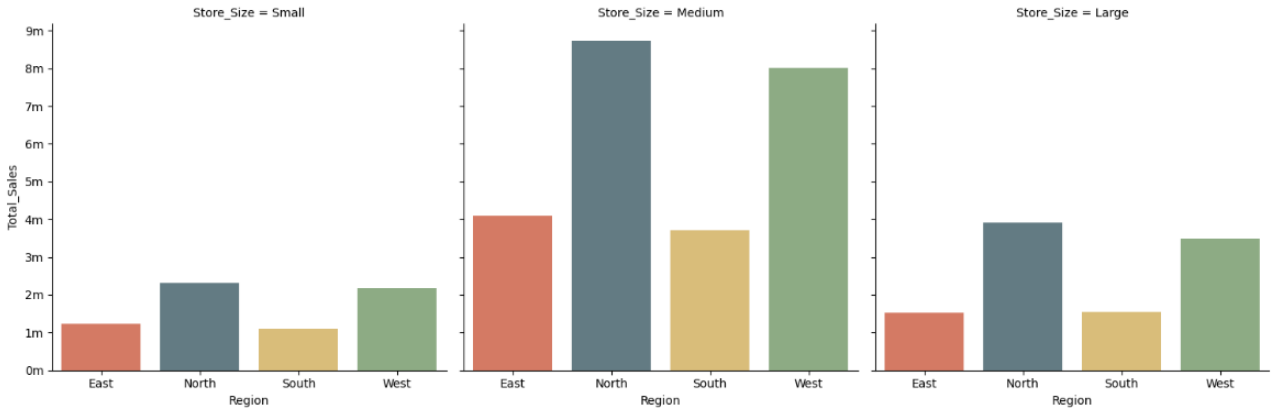

Figure 5 Total Sales by Region and Store Size

Figure 5 illustrates total sales broken down by region and store size. Sales contributions appear relatively balanced across store sizes within each region, with no single size category dominating disproportionately.

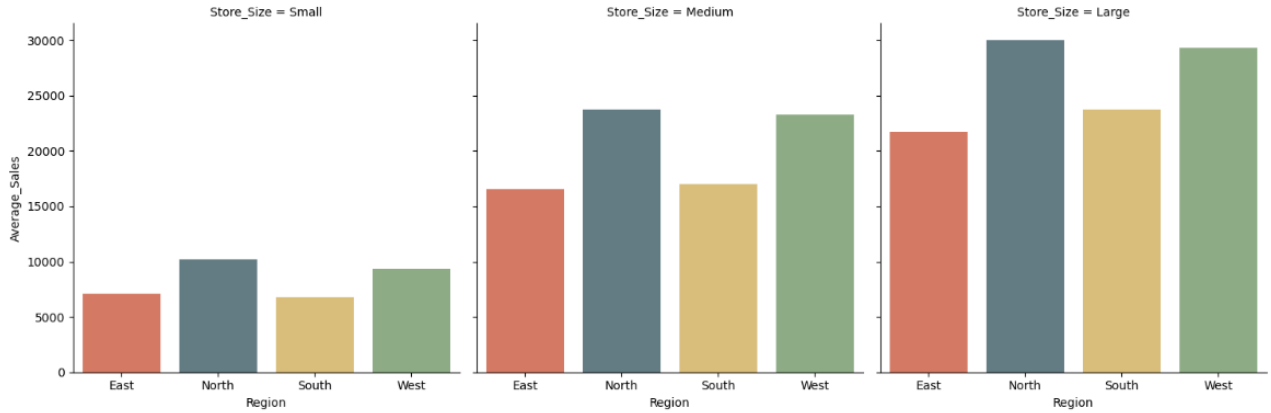

Figure 6 Mean Sales per Store by Region and Size

When comparing the averages in Figure 6, a clear pattern emerges: a larger store size directly increases average sales. Furthermore, the North and West regions remain at the top in terms of average sales, even when factoring in store size. For the best-performing stores, management could consider implementing an employee bonus program. However, it is also notable that the top store in the North has only €1,819 in sales per employee, which is significantly lower than other large stores.

2. PERFORMANCE DIFFERENCES WITHIN REGION AND STORE SIZE CATEGORIES

Figure 7 presents the best and the weakest performing store in each Region and Store Size category based on the combined performance score. The score is constructed from two metrics: Sales by Employee and Average Customer Sales. These specific metrics were selected because they evaluate both workforce productivity and sales effectiveness per visitor, providing a more balanced view of operational efficiency than total revenue alone. Both metrics were standardized before combining them, which means the score reflects relative performance compared to the overall average across all stores. A positive value indicates above-average performance, while a negative value indicates below-average performance.

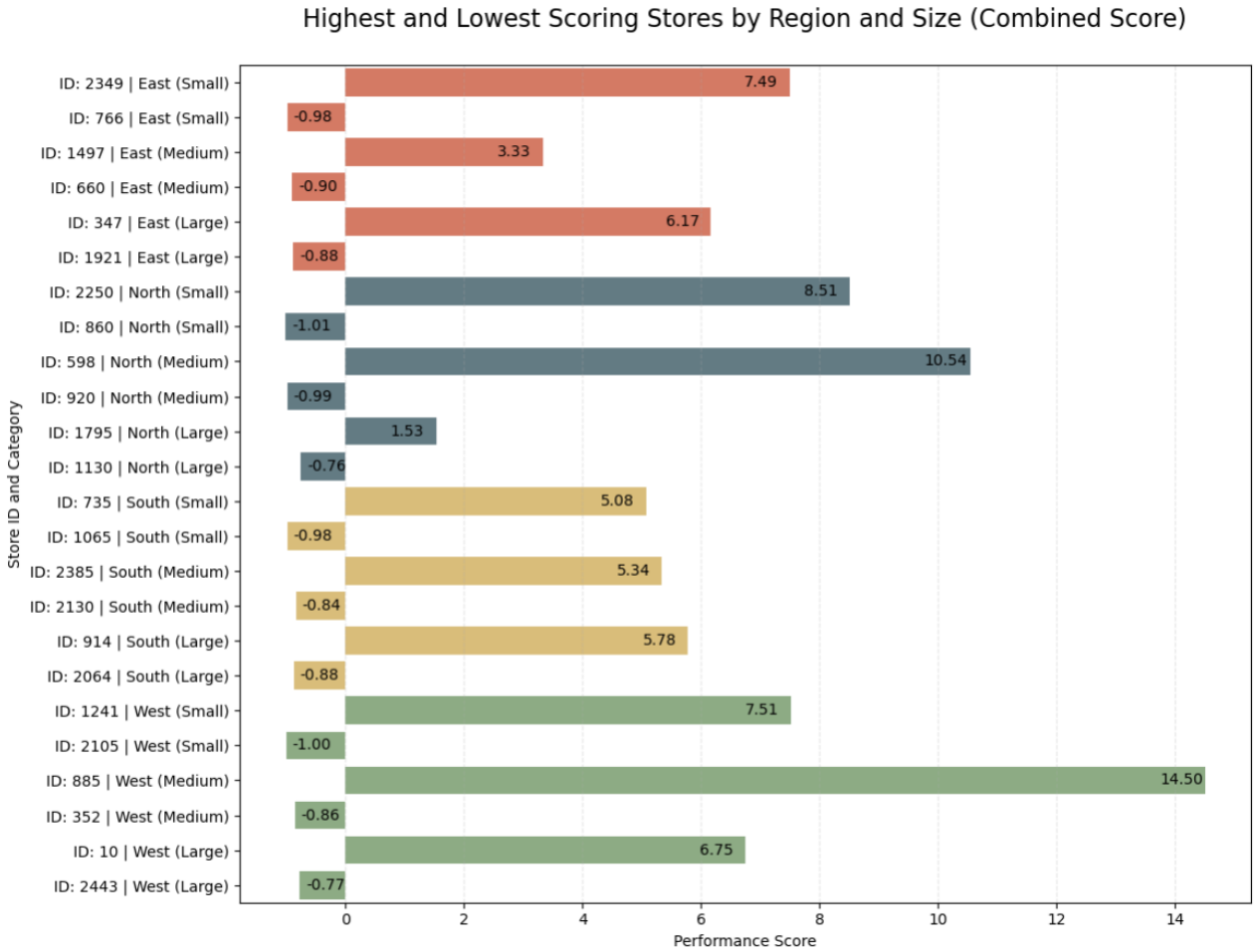

Figure 7 Comparison of the highest and lowest scoring stores by region and size.

By showing both the highest and lowest scoring store within each segment, Figure 7 highlights the performance gap inside the same operating environment. Stores in the same region and size category face broadly similar structural conditions, such as customer base and store format. Therefore, large differences in scores suggest that operational practices, management quality, or sales execution may vary significantly between stores.

In some categories, such as West Medium and North Medium, the best store performs above the overall average, while the weakest store in the same category is clearly below it. This indicates strong internal variation and significant potential improvement. In other categories, such as North Large and East Medium, even the top-performing store achieves only a relatively modest score compared to other segments. This may indicate structural challenges within those specific categories.

Overall, Figure 7 supports two managerial conclusions. First, top-performing stores within each segment can be used as benchmarks to identify effective practices. Second, large performance gaps within the same category suggest that there is room for internal learning and performance improvement without changing the overall business model or regional strategy.

3. MANAGERIAL RECOMMENDATIONS

Based on the exploratory analysis, the following actions are recommended to management:

Leverage Internal Benchmarking: As seen in Figure 7, there are large performance gaps between stores of the same size within the same region. The management should identify the operational practices of the top-performing stores (e.g., West Medium and North Medium) and implement them as training benchmarks for the underperforming stores.

Review the East and South Regions: Total sales and customer footfall are noticeably lower in the East and South compared to the North and West. The company should investigate whether this is due to market conditions, marketing efforts, or store visibility, and adjust the sales strategy accordingly.

Optimize Employee Incentives: Store size clearly drives total sales, but efficiency varies. Implementing performance-based bonuses tied to 'Sales by Employee' could motivate staff and increase overall profitability without needing to expand store areas.

4. LIMITATIONS

When interpreting these results, a few limitations apply. First, removing missing values reduced the original 2485 records to 2199. This 11.5 percent reduction was evenly distributed across regions, but introduces slight uncertainty to the exact totals. Second, the dataset only provides monthly sales and customer footfall. We lack data on profit margins, operational costs, and customer conversion rates. Future analyses would greatly benefit from actual cost data to determine true profitability rather than just top line revenue.